

PPSUV — Simulation Assignment

Step-by-Step Execution Guide

Power and Propulsion Systems for Unmanned Vehicles

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GT-Power | Simulink | ECMS

YOUR PERSONAL VALUES

Bore multiplier:	$\times 0.97$	$\rightarrow$	New Bore = Baseline $\times 0.97$	(−3%)
Stroke multiplier:	$\times 1.04$	$\rightarrow$	New Stroke = Baseline $\times 1.04$	(+4%)
PX vehicle config:	1			
ICE Torque multiplier:	$\times 1.02$	$\rightarrow$	Scale the ICE torque map by 1.02	(+2%)

IMPORTANT

These four numbers come from the assignment table next to **your** name. Do NOT use anyone else's values — they are different for every student.

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PART 1 — GT-Power: Engine Simulation

This part uses GT-Power to simulate a 4-cylinder turbocharged gasoline engine. You will change its size (bore and stroke), then run it under different conditions and record the results.

NOTE

GT-Power is a 1D engine simulation software. You open a model, change some numbers, click Run, then read results from plots. You do **NOT** need to build the engine from scratch — a template is already provided.

Step 1: Open the Template Model in GT-Power

Open GT-Power (GT-ISE) on your computer. Navigate to this file path:

```
GTI\v2016\examples\Engine_1D_Gas_Exchange_Combustion\Gasoline\
SI_4cyl_SITurb_PortFuelFilm_TWallSoln
```

Open the file SI_4cyl_SITurb_PortFuelFilm_TWallSoln.gtm (or .gti). This is the baseline 4-cylinder engine.

IMPORTANT

SAVE THIS FILE UNDER A NEW NAME IMMEDIATELY before changing anything.

Use: File → Save As → Harshal_Engine_Model.gtm
This way you never break the original template.

Step 2: Scale the Bore and Stroke (Your Values)

You need to change the physical dimensions of the engine cylinder:

Parameter	What it means	Baseline	Your multiplier	What to do
Bore	Cylinder diameter	~86 mm	$\times 0.97$	$86 \times 0.97 = 83.42$ mm
Stroke	Piston travel distance	~86 mm	$\times 1.04$	$86 \times 1.04 = 89.44$ mm
Intake valve $\varnothing$	Intake valve diameter	Varies	$\times 0.97$	Same as bore multiplier
Exhaust valve $\varnothing$	Exhaust valve diameter	Varies	$\times 0.97$	Same as bore multiplier

WHAT TO DO

In GT-Power:

1. Double-click the **Cylinder** component (usually a blue block labelled **EngineCylinder**). Look for fields named **Bore** and **Stroke**. Multiply each by your factor and type in the new values. Click OK.
2. Find the **intake and exhaust valve objects** (look for **IntakeValve** and **ExhaustValve** blocks). Find the **Diameter** or **MaxLift** field and multiply the original value by **0.97**.

TIP

The valve diameter scales with bore because valves must fit inside the cylinder: smaller bore → smaller valve. The actual baseline numbers are whatever you find in the template file — just multiply whatever is there by 0.97 (bore) or 1.04 (stroke).

Step 3: Set the Air-to-Fuel Ratio (AFR) to Stoichiometric

Stoichiometric means the perfect chemical balance of fuel and air — not too rich, not too lean.

AFR_{stoich} = 14.7 (for gasoline)

This means 14.7kg of air for every 1kg of fuel. It is called $\lambda = 1.0$.

WHAT TO DO

In the GT-Power model, find the fuel injector or combustion object. Look for a field called **Air-Fuel Ratio**, **Lambda**, or **Equivalence Ratio**. Set it to: **AFR = 14.7** OR **$\lambda = 1.0$** OR **$\varphi = 1.0$** . Make sure it is **FIXED** (not controlled by a lambda controller yet).

IMPORTANT

Do NOT turn on the lambda controller for Investigations 1, 2, 3. The AFR must be manually set for these. You will only control load via the throttle in Investigation 4.

Investigation 1 — Engine Speed Sweep

Run the engine at 5 different speeds and see how performance changes. This is the most basic investigation — do this first.

Conditions to set	Speeds to run (5 runs total)
AFR = 14.7 (stoichiometric, fixed)	Run 1: 1000 rpm
Throttle = Wide Open (WOT) = 100%	Run 2: 2000 rpm
Spark advance = baseline (no change)	Run 3: 3000 rpm
	Run 4: 4000 rpm
	Run 5: 5000 rpm

What to Extract and Plot (vs. Engine Speed)

After all 5 runs, make these plots with engine speed on the X-axis:

Plot	What it shows	Expected shape
BMEP vs speed	Brake Mean Effective Pressure	Bell curve — peaks mid-speed
IMEP vs speed	Indicated MEP (incl. friction)	Similar to BMEP, slightly higher
Brake Torque vs speed	Twisting force on crankshaft	Bell curve — peaks mid-speed
Brake Power vs speed	Actual power output (kW)	Rises then drops at high speed
Vol. Efficiency vs speed	How well air fills cylinder	Drops significantly at high rpm
BSFC vs speed	Fuel used per kWh of power	U-shape — best at mid-range
p-V diagram	Pressure vs volume inside cyl.	Closed loop — bigger = more work
p- θ diagram	Pressure vs crank angle	Peak pressure near TDC

Why the Results Look the Way They Do — Justification

- **BMEP and torque peak at mid-speed:** At low speed, not enough air enters quickly → less fuel burned → less force. At mid speed, air/fuel filling is ideal → maximum combustion pressure. At high speed, air cannot fill the cylinder fast enough (volumetric efficiency drops) → less force.
- **Brake Power keeps rising then falls:** $P = \text{Torque} \times \omega$. Even as torque drops, higher speed compensates... until torque drops too fast.
- **Volumetric efficiency drops at high rpm:** Air has inertia. At high speed there is not enough time to push air in before the valve closes → cylinder not fully filled.
- **BSFC is best at mid-range:** At low speed, friction losses are proportionally large → wastes fuel. At high speed, pumping losses increase → wastes fuel. In the middle: best balance of combustion efficiency and friction.

Investigation 2 — Air-to-Fuel Ratio (AFR) Variation

Repeat the full speed sweep (1000–5000 rpm) three times — once for each AFR setting.

Case	AFR Value	Lambda (λ)	What it means	Expected effect
Baseline	14.7	1.0	Stoichiometric — perfect balance	Reference (from Inv.1)
Rich (−20%)	11.76	0.8	Too much fuel — ‘rich mixture’	More torque, worse BSFC
Lean (+20%)	17.64	1.2	Too much air — ‘lean mixture’	Less torque, better BSFC

TIP

To set AFR −20%: multiply $14.7 \times 0.80 = 11.76$ (rich).

To set AFR +20%: multiply $14.7 \times 1.20 = 17.64$ (lean).

In the software, change the AFR field to these values and re-run the full speed sweep each time.

What to Plot

Plot the same 8 outputs as Investigation 1, but now put **ALL THREE** cases on the **SAME** graph so you can compare them. Use different colours or line styles.

Example: For BMEP vs Speed, you should have 3 lines: solid (AFR=14.7), dashed (AFR=11.76, rich), dotted (AFR=17.64, lean).

Why — Justification

- **RICH** ($\lambda < 1$): More energy available → higher BMEP and torque. But not all fuel can burn (not enough oxygen) → CO and HC emissions increase. BSFC worsens because you are wasting fuel that does not burn.
- **LEAN** ($\lambda > 1$): Less fuel → lower BMEP and torque. More oxygen available → cleaner combustion, lower CO/HC. BSFC improves (doing more with less fuel) — up to a limit. If too lean ($\lambda > 1.3$), combustion becomes unstable or misfires.
- **Volumetric efficiency does NOT change with AFR.** It only depends on how much *air* enters. AFR changes the fuel amount, not the air amount.

Investigation 3 — Spark Advance Variation

This investigation looks at *when* the spark fires inside the cylinder. Timing matters — too early or too late both waste energy.

NOTE

Spark advance = how many crank angle degrees **BEFORE** TDC the spark fires.
 Baseline = whatever value the template uses (e.g. 25° before TDC).
 You apply **CHANGES** (deltas) relative to that baseline — not absolute values.

Fixed conditions	Spark advance variations (5 runs)
Engine speed: pick one fixed speed (recommended: 3000 rpm) AFR = 14.7 (stoichiometric) Throttle = Wide Open (WOT)	Run 1: Baseline -10° (most retarded) Run 2: Baseline -5° Run 3: Baseline (no change) Run 4: Baseline $+5^\circ$ Run 5: Baseline $+10^\circ$ (most advanced)

Visual Explanation of Spark Timing

TOO RETARDED (-10°)	OPTIMAL (MBT)	TOO ADVANCED ($+10^\circ$)
Spark fires LATE — combustion happens during expansion. → Less work done on piston. → Hot exhaust gases (energy wasted as heat). → Lower BMEP, higher BSFC.	Spark fires at the PERFECT time. → Peak pressure near TDC. → Maximum work on piston. → Best BMEP, best BSFC.	Spark fires TOO EARLY — combustion fights the rising piston. → ‘Negative work’ — piston pushed back. → Risk of engine knock. → Lower BMEP, worse BSFC.

NOTE

MBT = Maximum Brake Torque timing. This is the ‘sweet spot’ for spark advance. Your baseline in the template is probably already close to MBT — that is why going either direction makes things worse.

Investigation 4 — Throttle Position (Part Load)

In Investigations 1–3 the throttle was fully open (Wide Open Throttle). Now you restrict the throttle to simulate driving at part load — like cruising in the city at low power.

TIP

You need to use a **DIFFERENT template** for throttle control. Import the ThrottleController from:
..\\GTI\\v2016\\examples\\Engine_1D_Gas_Exchange_Combustion\\Gasoline\\SI_4cyl_ThrottleController
Open this template and connect it to your scaled engine model.

Fixed conditions	Throttle targets (4 runs)
Fixed engine speed (e.g. 2000 or 3000 rpm)	Run 1: 50% of maximum torque
AFR = 14.7 (stoichiometric)	Run 2: 20% of maximum torque
Spark advance = baseline (no change)	Run 3: 10% of maximum torque
	Run 4: 5% of maximum torque

For this investigation, plot **BSFC against the load** (% of max torque) instead of against speed, plus the p-V and p- θ diagrams.

Why — Justification

- **PHYSICS:** As throttle closes, MAP (intake pressure) drops significantly. The engine has to ‘suck harder’ against the closed throttle → pumping losses increase. At 5% load, most of the fuel energy goes to overcome pumping losses — almost nothing useful comes out.
- **RESULT:** BSFC gets dramatically worse at part load. This is exactly why city driving (lots of low-load operation) burns more fuel per km than highway driving.
- **KEY INSIGHT:** This is the main reason why hybrid vehicles exist! A hybrid can turn off the ICE at low load and use the electric motor instead. When the ICE does run, it operates at higher load (better BSFC region).

PART 2 — Simulink/ECMS: Vehicle Simulation

This part simulates a complete plug-in hybrid electric vehicle (PHEV) driving along standardised test cycles. The ECMS (Equivalent Consumption Minimisation Strategy) decides at every instant whether to use the engine or the battery.

NOTE

You are NOT building the vehicle model from scratch. You open the PX vehicle model (config 1 — your assignment) in Simulink and change a few numbers. Simulink is a graphical simulation environment from MATLAB. Models look like connected blocks. ECMS is the energy management strategy — it is already implemented in the model. You just tune one number: the **Equivalence Factor (EF)**.

Step 1: Open the PX Vehicle Model (Configuration 1)

Open MATLAB/Simulink. Navigate to the PX vehicle model for configuration 1 (your PX = 1).

WHAT TO DO

The PX model should be provided by your professor or in the course materials folder. Look for a file called something like `PX_vehicle_config1.slx` or similar. Open it in Simulink by double-clicking or using File → Open in MATLAB.

Scale the ICE Torque Map (Your Value: $\times 1.02$)

Inside the model, find the ICE (Internal Combustion Engine) block. It contains a lookup table (map) that tells the engine how much torque it can produce at each speed. You need to multiply **ALL** values in this map by 1.02.

`New_Torque_Map = Original_Torque_Map  $\times$  1.02`

This represents a slightly more powerful engine for your individual assignment (+2% maximum torque).

WHAT TO DO

1. Double-click the ICE block in Simulink.
2. Find the torque map parameter (usually a matrix/array called `TorqueMap` or `MaxTorque`).
3. In the MATLAB command window, type: `TorqueMap = TorqueMap * 1.02`
4. Or edit the matrix directly in the block dialog and multiply each value by 1.02.

Step 2: Run WLTC Class 3 with Constant Equivalence Factor (EF)

WLTC Class 3 is the standard driving cycle for heavier vehicles. It has urban, rural, and motorway sections.

NOTE

ECMS = Equivalent Consumption Minimisation Strategy.

At every time step, ECMS decides: ‘Should I use the ICE or the battery right now?’

It does this by assigning a ‘cost’ to using the battery, equivalent to burning fuel.

The Equivalence Factor (EF) = s = this cost. If s is high $\rightarrow$ battery is expensive to use $\rightarrow$ ICE runs more.

$$H(t) = \dot{m}_{\text{fuel}}(P_{\text{ice}}) + s \times P_{\text{bat}}/\text{LHV}$$

ECMS minimises H at every instant. s is the equivalence factor you must tune.

How to Tune the EF (Trial and Error)

- Step 1:** Run the simulation with $s = 2.5$ (start here). Run the full WLTC Class 3 cycle.
- Step 2:** At the end, check the SOC plot. Is $\text{SOC}_{\text{final}} \approx \text{SOC}_{\text{initial}}$? (e.g. both $\sim 50\%$?)
- Step 3:** If $\text{SOC}_{\text{final}} > \text{SOC}_{\text{initial}}$ $\rightarrow$ battery overcharged $\rightarrow$ **DECREASE** s (try $s = 2.0$).
- Step 4:** If $\text{SOC}_{\text{final}} < \text{SOC}_{\text{initial}}$ $\rightarrow$ battery drained $\rightarrow$ **INCREASE** s (try $s = 3.0$).
- Step 5:** Repeat until $|\text{SOC}_{\text{final}} - \text{SOC}_{\text{initial}}| < 1\%$. This is your tuned constant EF.

SOC goes UP	SOC BALANCED ✓	SOC goes DOWN
Battery overcharged $\rightarrow$ ICE running too much $\rightarrow$ DECREASE s	$\text{SOC}_{\text{end}} \approx \text{SOC}_{\text{start}}$ $\rightarrow$ Correct EF found!	Battery drained $\rightarrow$ Battery used too much $\rightarrow$ INCREASE s

What to Extract and Plot After Tuning

Signal to Plot	Units	Where to find it in Simulink
SOC vs time	%	Battery block output or scope named ‘SOC’
ICE Power vs time	kW	ICE block output ‘Power’ or ‘P_ice’
ICE Speed vs time	rpm	ICE block output ‘Speed’ or ‘n_ice’
ICE Torque vs time	Nm	ICE block output ‘Torque’ or ‘T_ice’
EM Power vs time	kW	Electric motor block output ‘Power’
EM Speed vs time	rpm	Electric motor block output ‘Speed’
EM Torque vs time	Nm	Electric motor block output ‘Torque’
CO ₂ emissions	g/km	Emissions block or post-processing script
CO emissions	g/km	Emissions block
NOx emissions	g/km	Emissions block
HC emissions	g/km	Emissions block
Fuel economy	l/100 km	Fuel consumption block

Step 3: Run Adaptive EF (Variable Equivalence Factor)

Now switch the model to ‘adaptive ECMS’ mode. The EF is no longer constant — it changes based on the current SOC.

NOTE

The idea: if the battery is getting low, increase s to make the engine run more and recharge. If the battery is getting too full, decrease s to use more electricity. This keeps the SOC closer to the target (usually 50%) throughout the cycle.

WHAT TO DO

1. Find the ECMS block and look for an ‘Adaptive’ switch or parameter.
2. Enable adaptive mode (set flag to 1 or ‘true’).
3. Use your tuned constant EF as the starting value.
4. Re-run the WLTC Class 3 cycle.

Compare the adaptive EF results against the constant EF results from Step 2. You should see the SOC trace is smoother and stays closer to the target.

Step 4: Run a Different Driving Cycle

WITHOUT changing any ECMS settings (keep the same EF values), run the model on a completely different driving cycle. This shows how the same vehicle behaves in different real-world conditions.

NOTE

Pick ONE cycle from the list. Recommended: **FTP75** (lots of stop-and-go, good contrast with WLTC) or **Artemis Motorway 130** (high speed, ICE-dominant — very different behaviour).

Cycle	Type	Avg Speed	Best for showing...
FTP75	USA urban	~31 km/h	Stop-and-go, lots of regen braking
Artemis Rural Road	European rural	~57 km/h	Mixed conditions
Artemis Motorway 130	European motorway	~108 km/h	High speed, ICE dominates
JC08 Hot	Japanese urban	~25 km/h	Slow urban driving
UDDS	USA urban	~31 km/h	Similar to FTP75 first phase
LA92	USA aggressive	—	More aggressive than FTP75

Why the Same EF Gives Different Results on Different Cycles

- The EF was tuned for WLTC. On a different cycle, the energy demand pattern is different.
- FTP75 has more stop-and-go → more regenerative braking → SOC tends to increase.
- Motorway cycle has sustained high speed → ICE runs more → battery not recharged as much.
- **RESULT:** $\Delta\text{SOC} \neq 0$ on the new cycle — this is expected and shows why adaptive EF is important.

Step 5: Sensitivity Analysis: $\pm 25\%$ Vehicle Parameter Variation

Without changing any ECMS settings, vary at least ONE vehicle parameter by $+25\%$ and -25% . This shows how vehicle design affects performance.

Parameter	Where to change in Simulink	Effect of increasing
Vehicle mass (kg)	Vehicle block $\rightarrow$ 'Mass' parameter	More energy to accelerate $\rightarrow$ worse economy
Frontal area A_f (m^2)	Aerodynamics block $\rightarrow$ 'Af' parameter	More aero drag $\rightarrow$ worse at high speed
Battery capacity (kWh)	Battery block $\rightarrow$ 'Capacity' or 'Q_max'	More EV range $\rightarrow$ more ECMS flexibility

TIP

RECOMMENDED: vary vehicle mass. It has the largest effect and is easiest to justify physically.

Run 3 simulations: Mass $\times 0.75$ (light), Mass $\times 1.0$ (baseline), Mass $\times 1.25$ (heavy).

Physical Justification

- **MASS:** Heavier vehicle needs more tractive force $\rightarrow$ more fuel burned per km (rolling resistance and inertia both increase).
 - Rolling resistance: $F_r = c_r \times m \times g \rightarrow$ more mass = more F_r .
 - Kinetic energy: $E_k = \frac{1}{2}mv^2 \rightarrow$ more mass = more energy needed to accelerate.
 - But: more mass also means more energy available during regenerative braking.
- **FRONTAL AREA:** Bigger frontal area increases aerodynamic drag proportionally with v^2 .
 - $F_a = \frac{1}{2}\rho \times A_f \times c_x \times v^2 \rightarrow$ doubling A_f doubles the aerodynamic force.
 - Effect is most severe at motorway speeds (v^2). Negligible at low urban speeds.
- **BATTERY:** Larger battery gives more flexibility to the ECMS to defer ICE use.
 - ECMS can run in EV mode for longer periods $\rightarrow$ ICE can be operated at better efficiency points.
 - However: larger battery = heavier vehicle $\rightarrow$ partially cancels the benefit.

What to Compare Across All Configurations

- ICE and EM operating points overlaid on BSFC and efficiency maps.
- Fuel economy (l/100 km) and CO_2 emissions.
- SOC trajectory over the cycle.
- Instantaneous power split between ICE and EM.

WHAT TO DO

For the BSFC map plot: scatter the (rpm, torque) operating points of the ICE on top of the colour map. Points clustered in the green/low BSFC island = good. Points in the red zone = inefficient operation. A heavier vehicle should show points shifted to higher torque (more load needed) — possibly into better efficiency regions.

Emissions Comparison — EURO 6e and EU CO₂ Target

Compare your simulated emissions to the regulatory limits. This is required in the report.

Pollutant	EURO 6e Limit (gasoline)	Your simulated value	Pass / Fail?
CO	1.000 g/km	_____	Compare
NO _x	0.060 g/km	_____	Compare
HC	0.100 g/km	_____	Compare
HC + NO _x combined	0.170 g/km	_____	Compare
CO ₂	81 g/km (2025 EU target)	_____	Compare

NOTE

EURO 6e limits apply at the type approval level — measured on the WLTC cycle.
 The EU CO₂ target of 81 g/km is a fleet average for new passenger cars by 2025.
 A plug-in hybrid running in charge-depleting mode can achieve very low CO₂ per km (electric driving counts as zero emissions in the official test).

Final Checklist — Before You Submit

Part 1 — GT-Power Checklist

- ☐ Bore scaled: $\text{Baseline} \times 0.97$
- ☐ Stroke scaled: $\text{Baseline} \times 1.04$
- ☐ Intake and exhaust valve diameters scaled: $\text{Baseline} \times 0.97$
- ☐ AFR set to 14.7 (stoichiometric) and FIXED for Investigations 1, 2, 3
- ☐ Investigation 1: Speed sweep (1000–5000 rpm) completed — 5 runs
- ☐ Investigation 1: BMEP, IMEP, Torque, Power, Vol. Efficiency, BSFC, p-V, p- θ plotted
- ☐ Investigation 2: AFR sweep (11.76, 14.7, 17.64) at all 5 speeds — 15 runs total
- ☐ Investigation 2: All plots show 3 overlaid lines (lean / stoich / rich)
- ☐ Investigation 3: Spark advance variations (−10, −5, 0, +5, +10 CAD) at fixed speed — 5 runs
- ☐ Investigation 3: All plots vs spark advance generated
- ☐ Investigation 4: Throttle sweep (50%, 20%, 10%, 5%) at fixed speed — 4 runs
- ☐ Investigation 4: BSFC vs load plotted, p-V and p- θ diagrams included
- ☐ Physical justification written for each investigation
- ☐ All plots clearly labelled with axes, units, and legend

Part 2 — Simulink/ECMS Checklist

- ☐ PX vehicle configuration 1 opened
- ☐ ICE torque map multiplied by 1.02
- ☐ WLTC Class 3 run with constant EF — EF tuned until $\Delta\text{SOC} < 1\%$
- ☐ Adaptive EF run on WLTC Class 3
- ☐ Constant EF vs Adaptive EF results compared
- ☐ Second driving cycle chosen and simulated (no ECMS changes)
- ☐ SOC, ICE/EM power/speed/torque plotted vs time for all runs
- ☐ CO₂, CO, NO_x, HC emissions extracted (g/km)
- ☐ Fuel economy extracted (l/100 km)
- ☐ ICE operating points plotted on BSFC map
- ☐ EM operating points plotted on efficiency map
- ☐ Emissions compared against EURO 6e table
- ☐ CO₂ compared against EU 2025 target (81 g/km)
- ☐ Sensitivity analysis: $\pm 25\%$ on at least one vehicle parameter
- ☐ All three configurations compared (baseline, +25%, −25%)
- ☐ Operating point maps compared across configurations
- ☐ Physical justification written for all variations

IMPORTANT

Double-check that your bore, stroke, and ICE torque values are from **YOUR** row in the assignment table.

Bore = 0.97, Stroke = 1.04, PX config = 1, ICE Torque mult = 1.02.

If these are wrong, all your results will be wrong.

Key Equations Reference

Use these in your justifications. You do not need to derive them — just state them and explain what each term means.

$$\text{BMEP} = \frac{W_{\text{indicated}} \times 2}{V_{\text{displacement}}} \quad [\text{Pa}]$$

W = work per cycle, V = cylinder swept volume, factor 2 because 4-stroke fires every 2 revolutions.

$$P_{\text{brake}} = \text{Torque} \times \omega = \text{Torque} \times \frac{2\pi n}{60} \quad [\text{W}]$$

P = brake power, n = engine speed in rpm, ω = angular velocity in rad/s.

$$\eta_{\text{vol}} = \frac{\dot{m}_{\text{air,actual}}}{\rho_{\text{air,ref}} \times V_{\text{displacement}}} \quad [-]$$

Volumetric efficiency — how much air the cylinder actually traps vs its maximum possible.

$$\text{BSFC} = \frac{\dot{m}_{\text{fuel}}}{P_{\text{brake}}} \quad [\text{g/kWh}]$$

Brake Specific Fuel Consumption — lower is better. Typical good value: 250–280 g/kWh.

$$F_{\text{aero}} = \frac{1}{2} \rho_{\text{air}} \times A_f \times c_x \times v^2 \quad [\text{N}]$$

Aerodynamic drag force — grows with v^2 so it dominates at motorway speeds.

$$F_{\text{roll}} = c_r \times m_{\text{vehicle}} \times g \quad [\text{N}]$$

Rolling resistance — proportional to vehicle mass and gravity, roughly constant with speed.

$$H(t) = \dot{m}_{\text{fuel}}(P_{\text{ice}}) + s \times \frac{P_{\text{bat}}}{\text{LHV}} \rightarrow \text{minimise} \quad [\text{kg/s}]$$

ECMS Hamiltonian — at each time step, ECMS chooses P_{ice} to minimise this. s = equivalence factor.

$$\frac{d\text{SOC}}{dt} = -\frac{P_{\text{bat}}}{Q_{\text{bat}} \times V_{\text{oc}}} \quad [1/\text{s}]$$

SOC dynamics — P_{bat} = battery power (negative = charging), Q_{bat} = capacity in Ah, V_{oc} = open circuit voltage.